The Chief Data Officer's guide to digital transformation

How to push your business's data operations to the forefront of innovation and data-driven value

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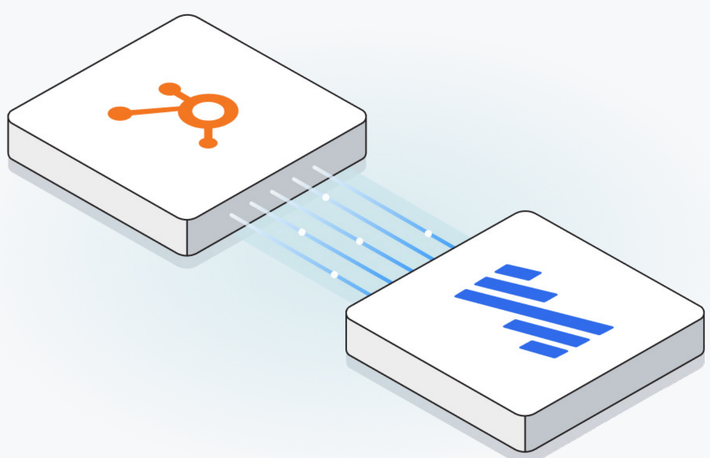
Executive summary

Chief Data Officers (CDOs) have a mandate to bring digital transformation to their organizations. In the context of data operations, digital transformation means moving from crude or reactive uses of data, such as descriptive analytics, to higher-value, more sophisticated and innovative uses of data, such as predictive analytics.

Chief Data Officers face a number of **well-known challenges** such as institutional inertia and a lack of clear expectations. Moreover, CDOs must contend with the following:

- Teams must make high-stakes decisions that are best supported with data rather than educated guesswork. The more agile the company, the better, but decentralizing decision making carries its own risks. To compound these difficulties, teams are often partly or wholly remote.
- Competition and macroeconomic headwinds mean that organizations must always find ways to do more with less in terms of headcount and budget.
- As the volume of data grows, so does the complexity of its handling. Governments write increasingly stringent regulations, while consumers simultaneously expect greater personalization and stronger security.

These challenges contribute to the notoriously high turnover of CDOs, with an average tenure of fewer than **30 months**. The clock for achieving transformational change is always ticking. The goal of this guide is to help CDOs identify critical junctures on their roadmaps to bring their organizations to the forefront of data maturity and innovation, and in the process beat the clock.



There are several stages that a roadmap for digital transformation must progress through. These four stages – data centralization, infrastructure modernization, data democratization and building data solutions – not only constitute a progression but also represent use cases that have a common technological solution.

This solution is the automated data movement platform.

1. **Data centralization** is the classical data integration problem. As SaaS applications for every business operation proliferate, organizations must consolidate data from a growing variety of sources onto a platform that decision makers can easily and performantly access. Data on a single, centralized platform produces a full view of an organization's operations, customers and products, enabling analytics and all further uses of data. Unlike federated analytics, it can also perform at very large scales. However, it is a deceptively complex engineering challenge that is best solved with the use of a fully managed, automated data movement platform.

2. Once data is centralized, **data democratization** brings accessibility and governance to stakeholders and decision makers across an organization. Distributing access to data and decision-making authority enables greater organizational agility and better responsiveness to dynamic, changing markets. However, exposing data to a larger audience also carries security risks that must be addressed using security features and governance tools.

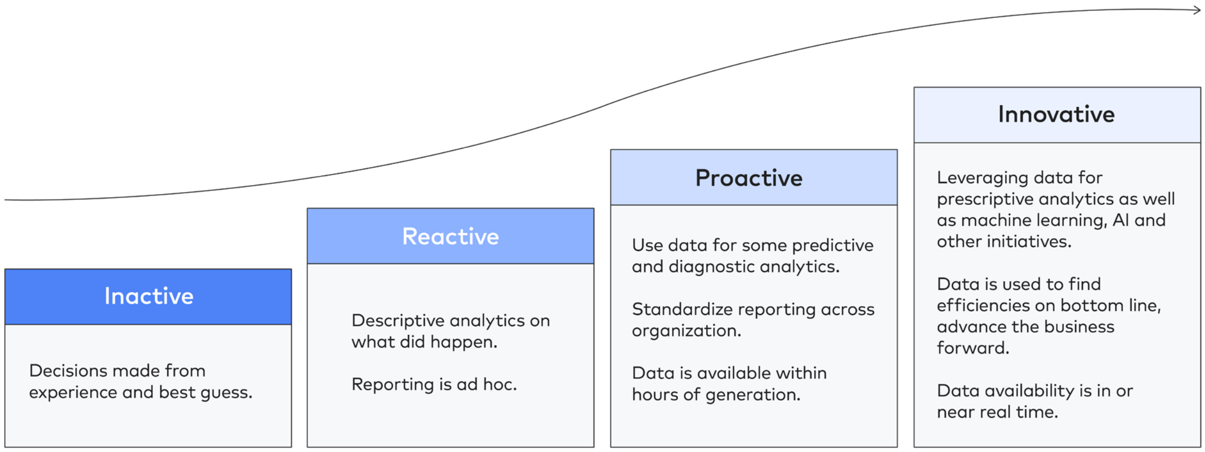
3. **Building data solutions** consists of finding ways to share data internally or with third parties, turn it into products or otherwise monetize it. Raw data, insights and systems fueled by data such as predictive models or artificially intelligent agents are all valuable commodities. Such pursuits require data sharing and extensibility features.

As an organization progresses through the stages listed above, it will also be incentivized to **modernize its infrastructure**, adding and removing tools and platforms to meet emerging needs. A common example is moving from on-premise to cloud or hybrid infrastructure. This is an incremental, continuous progress. Ripping and replacing is rarely practical because of existing commitments.

All roadmaps to digital transformation must begin with data centralization. From there, data democratization is a logical next step. Mastery over data centralization also enables an organization to monetize data by building data solutions. All throughout, an organization should continuously evolve its data infrastructure in response to changing needs.

A structured approach to data

Data is the most essential resource for organizations of all kinds, used to support decisions, automate processes and power innovative products. Its value increases with freshness and scale. You can mentally model the use of data as a progression of four steps called the data maturity curve:



Data-driven companies that use data in proactive and innovative ways are far more likely to thrive than companies that rely more on guesswork and ad hoc reporting.

Organizations today use a wide range of systems that produce digital footprints, ranging from applications and databases to event streams and files. All of these digital footprints serve as clues and building blocks to valuable insights. In order to lead and unify the efforts involved in gaining visibility and control over the growing volume, velocity and variety of data created by business activities, many organizations have created the position of the Chief Data Officer (CDO).